Experimental Study of Phase Equilibria and Thermodynamic Optimization of the Fe-Zn-O System

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The Fe-Zn-O phase diagram in air was studied over the temperature range from 900 °C to 1500 °C. The compositions of the phases in quenched samples were obtained by electron probe X-ray microanalysis (EPMA). This experimental technique is not affected by zinc losses resulting from vaporization of zinc at high temperatures. The model for the spinel solid solution was developed within the framework of the compound-energy formalism (CEF). The choice of parameters of the CEF and the sequence of their optimization can have a major influence on the predictions in multicomponent phases. These choices can only be made rationally by reference to the specific model being represented in the CEF. This is discussed for the case of the two-sublattice spinel model. In the limiting case, the proposed model reduces to the model by O'Neill and Navrotsky for spinels. When the CEF is used in combination with the equation of Hillert and Jarl to describe the magnetic contribution to thermodynamic functions of a solution, it is necessary to assign certain values of magnetic properties to all pseudocomponents and to magnetic interaction parameters to obtain the most reasonable approximation of the magnetic properties of a solution. It was shown how this can be done based on very limited experimental data. The same equations can be used when the Murnaghan or the Birch–Murnaghan equation is combined with the CEF to describe the pressure dependence of thermodynamic functions. The polynomial model was used to describe the properties of wustite and zincite, and the modified quasichemical model was used for the liquid slag. All thermodynamic and phase-equilibria data on the Fe-O and Fe-Zn-O systems were critically evaluated, and parameters of the models were optimized to give a self-consistent set of thermodynamic functions of the phases in these systems. All experimental data are reproduced within experimental error limits. These include the thermodynamic properties of phases (such as specific heat, heat content, entropy, enthalpy, and Gibbs energy); the cation distribution between octahedral and tetrahedral sites in spinel; the oxygen partial pressure over single-phase, two-phase, and three-phase regions; the phase boundaries (liquidus, solidus, and subsolidus); and the tie-lines.

I. INTRODUCTION

THE study of the Fe-Zn-O system reported here is a part of a wider research program, which combines experimental measurements and thermodynamic modeling to characterize lead/zinc industrial slags and sinters. This program has resulted in the development of a self-consistent thermodynamic database which can be used with the F*A*C*T thermodynamic software package[1] to predict phase relations in the six-component system PbO-ZnO-CaO-FeO-Fe_2O_3-SiO_2 and to calculate various slag properties, including activities and liquidus temperatures over a wide range of compositions and temperatures for lead/zinc industrial smelting processes. Some other subsystems studied within this overall research program have already been described in previous publications by the present authors.[2–8]

In the thermodynamic "optimization" of a system, all available thermodynamic and phase-equilibrium data are evaluated simultaneously in order to obtain one set of model equations for the Gibbs energies of all phases as functions of temperature and composition. From these equations, all of the thermodynamic properties and the phase diagrams can be back-calculated. In this way, all the data are rendered self-consistent and consistent with thermodynamic principles. Thermodynamic property data, such as activity data, can aid in the evaluation of the phase diagram, and phase-diagram measurements can be used to deduce thermodynamic properties. Discrepancies in the available data can often be resolved, and interpolations and extrapolations can be made in a thermodynamically correct manner.

The present study started with an initial critical evaluation of all available reliable phase diagrams and thermodynamic data for the Fe-Zn-O system. The phase equilibria among hematite, spinel, zincite, and wustite solid solutions have been investigated extensively (the generic name "spinel" is used in this article for the solid solution between zinc ferrite and magnetite). However, significant discrepancies between experimental data were revealed, which resulted from experimental difficulties caused by high equilibrium zinc pressures at elevated temperatures. The high vaporization rates of zinc tend to change the sample composition during experiments. To resolve these problems, a new experimental investigation was carried out using an experimental technique which involves quenching of equilibrated samples followed by electron probe X-ray microanalysis (EPMA). This method eliminates the errors caused by changes in the sample bulk composition and makes it possible to characterize the phase equilibria in the Fe-Zn-O system with higher accuracy.

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The experimental measurements in air are described in this article. The phase equilibria between wustite and zincite at iron saturation have been reported earlier.[9]

The new experimental information and the literature data on the Fe-Zn-O system were optimized to obtain parameters of the thermodynamic models for the liquid and solid phases in this system. For the molten slag, a modified quasi-chemical model was used. This model has been described in several publications.[10–13] A model based on the compound-energy formalism (CEF) was developed for the spinel, and the polynomial model was used for the wustite and zincite solid solutions. The calculations were performed with the F*A*C*T thermodynamic computing system.[1]

The CEF is sometimes called the "compound-energy model." However, it is not a model, but rather a general formalism to which many sublattice models can be reduced. For this reason, the parameters of the formalism can have different physical meanings for different models. The choice of parameters and their sequence of optimization can have a major influence upon the predictions in multicomponent phases, and these choices can only be made rationally by reference to the specific model. This fact is not generally appreciated and, as a result, the CEF is sometimes applied incorrectly. In this article, the rational choice of parameters and their sequence of optimization is proposed for the two-sublattice spinel model.

II. EXPERIMENTAL

A. *General Description*

The experimental procedure was similar to that described previously by the present authors.[2,7,8] Oxide powders of ZnO and Fe_2O_3 of 99.5+ wt pct purity were used as starting materials. Pure powders were calcined at 110 °C, weighed in desired proportions, and mixed thoroughly in an agate mortar. Mixture compositions were chosen so as to obtain equilibria between two coexisting condensed phases. The mixtures were pelletized and equilibrated in platinum crucibles in air for times from 18 hours to 10 days.

The temperature was controlled to within ± 1 K. The working thermocouple was positioned next to the sample and periodically tested against a calibrated thermocouple. The overall temperature accuracy was estimated to be ± 5 K.

Sample weights were monitored before and after the experiments. All weight measurements were performed on an analytical balance with an accuracy of ± 0.1 mg. After final quenching into iced water or air, samples were mounted and polished. Microstructures were examined in detail using optical as well as scanning electron microscopy coupled with energy-dispersive spectra (EDS) analysis.

The EPMA of phase compositions was carried out with a JEOL* 8800L electron probe microanalyzer with wavelength-dispersive detectors. An accelerating voltage of 15 kV and a probe current of 15 nA were used. Zn_2SiO_4* and Fe_2O_3* were used as standards for EPMA measurements of zinc and iron concentrations, respectively. The Duncumb–Philibert ZAF correction procedure supplied with the JEOL-8800L was applied. The average accuracy of the EPMA measurements is within ± 1 wt pct.

*JEOL is a trademark of Japan Electron Optics Ltd., Tokyo.

*The Zn_2SiO_4 powder was supplied by Micro-Analysis Consultants Ltd., Cambridge, United Kingdom.

*The Fe_2O_3 powder was supplied by Charles M. Taylor, Co., Stanford, CA.

An X-ray powder diffraction analysis was performed in some cases to confirm the phase identification. It was carried out with a PHILIPS* PW1130 X-ray diffractometer with a graphite monochromator, using Cu K_α radiation.

*PHILIPS is a trademark of Philips Electronic Instruments Corp., Mahwah, NJ.

Attainment of equilibrium was checked by comparison of the samples equilibrated for different times. In order to confirm the achievement of equilibrium, the equilibration experiments were repeated starting from both pure oxide powders and from single-phase solid solutions.

On quenching, the phase assemblage that exists at the equilibration temperature is frozen in the sample. The quenched samples were mounted and polished; then, the compositions of the phases were measured with the electron probe, thereby giving the subsolidus compositions at the equilibration temperature.

This experimental technique was shown to be more efficient, more accurate, and more reliable than conventional techniques for phase-diagram determination.[2,7,8] The ability of the EPMA to deliver subsolidus data is of particular importance for the investigated system, since a number of compounds form extensive solid solutions. The changes of the sample composition during experiments, which result from the vaporization of zinc, do not affect the final accuracy of the results, since the EPMA measurement is carried out after the equilibration and quenching. Since zinc losses do not affect accuracy when this technique is used, it was possible to increase the equilibration time significantly. These features of the experimental technique are of primary importance for the study of the Fe-Zn-O system.

B. *Experimental Results for the Fe-Zn-O System in Air*

The compositions of the phases from the samples where a spinel + hematite or spinel + zincite equilibrium was obtained are reported in Table I. The following data for each experiment are given in the table: the experiment number (column 1), temperature (column 2), equilibration time (column 3), quenching medium (column 4), initial bulk composition (columns 5 and 6), phases in equilibrium (column 7), and compositions of condensed phases obtained by EPMA measurements (columns 8 and 9).

The EPMA provides only the total metal content. No ferrous/ferric analysis was carried out. The compositions of solid solutions, therefore, are given as $Zn^{2+}/(Zn^{2+} + Fe^{2+} + Fe^{3+})$ and $(Fe^{2+} + Fe^{3+})/(Zn^{2+} + Fe^{2+} + Fe^{3+})$ mole ratios.

The achievement of equilibrium was confirmed by the results of experiments with different sintering times. The fact that no further change in spinel composition occurred as the sintering time was increased indicated the achievement of equilibrium (*e.g.*, experiments 7 through 9). The achievement of equilibrium was also confirmed by approaching the equilibrium starting from a mixture of pure ZnO and Fe_2O_3

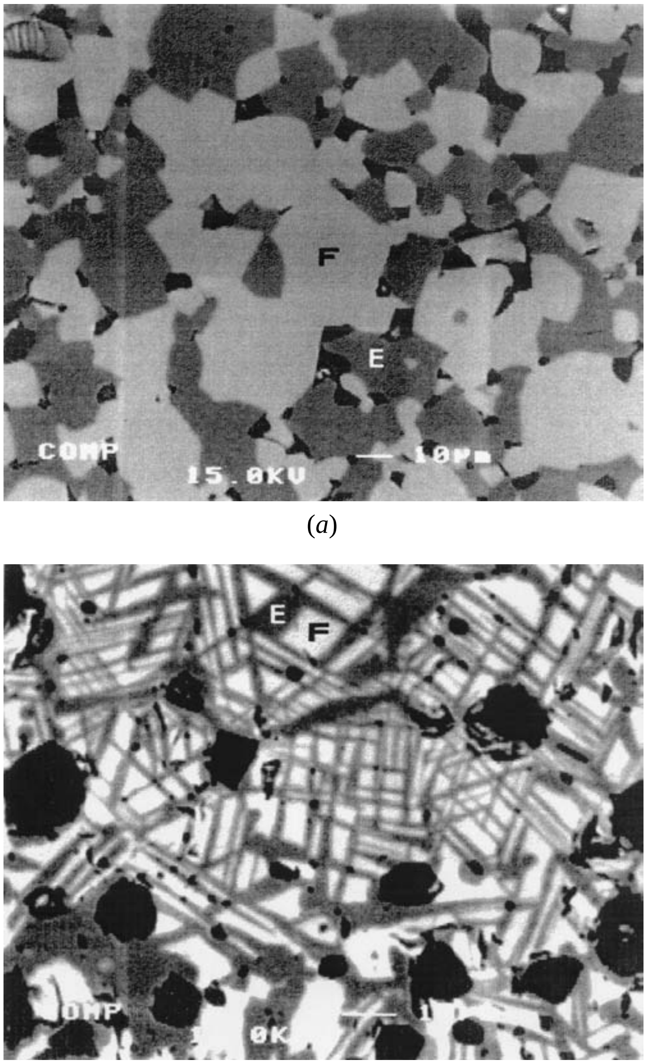
Table I. Results of Experiments on the System Fe-Zn-O in Equilibrium with Air

Experiment	Temperature (°C)	Equilibrium Time (h)	Quenching Medium	Initial Composition: ZnO (wt pct)	Initial Composition: Fe_2O_3 (wt pct)	Phases in Equilibrium	Mole Fraction: (Zn^{2+})	Mole Fraction: $(Fe^{2+} + Fe^{3+})$
1	2	3	4	5	6	7	8	9
1	900	174	air flow	14.5	85.5	Hem	0.008	0.992
						Sp	0.312	0.688
2	900	235	air flow	25.0	75.0	Hem	0.008	0.992
						Sp	0.315	0.685
3	1000	174	air flow	14.5	85.5	Hem	0.004	0.996
						Sp	0.296	0.704
4	1000	174	air flow	Sp from experiment 15		Hem	0.004	0.996
						Sp	0.295	0.705
5	1000	212	iced water	25.0	75.0	Hem	0.005	0.995
						Sp	0.291	0.709
6	1000	360	air flow	25.0	75.0	Hem	0.011	0.989
						Sp	0.297	0.703
7	1100	91	air flow	25.0	75.0	Hem	0.006	0.994
						Sp	0.266	0.734
8	1100	168	air flow	14.5	85.5	Hem	0.005	0.995
						Sp	0.267	0.733
9	1100	264	air flow	25.0	75.0	Hem	0.010	0.990
						Sp	0.272	0.728
10	1200	19	air flow	14.5	85.5	Hem	0.003	0.997
						Sp	0.217	0.783
11	1200	96	air flow	14.5	85.5	Hem	0.003	0.997
						Sp	0.215	0.785
12	1200	113	air flow	14.5	85.5	Hem	0.003	0.997
						Sp	0.219	0.781
13	1300	22	air flow	8.0	92.0	Hem	0.001	0.999
						Sp	0.124	0.876
14	1350	18	iced water	14.5	85.5	Sp	0.137	0.863
15	1350	24	air flow	14.5	85.5	Sp	0.137	0.863
16	1350	25	air flow	2.0	98.0	Hem	0.001	0.999
						Sp	0.058	0.942
17	1450	18	iced water	14.5	85.5	Sp	0.129	0.871
18	900	212	air flow	50.0	50.0	Z	0.950	0.050
						Sp	0.331	0.669
19	900	235	air flow	50.0	50.0	Z	0.958	0.042
						Sp	0.326	0.674
20	1000	212	air flow	50.0	50.0	Z	0.953	0.047
						Sp	0.328	0.672
21	1100	91	air flow	50.0	50.0	Z	0.954	0.046
						Sp	0.328	0.672
22	1100	168	air flow	50.0	50.0	Z	0.957	0.043
						Sp	0.329	0.671
23	1200	91	air flow	50.0	50.0	Z	0.953	0.047
						Sp	0.325	0.675
24	1200	113	air flow	50.0	50.0	Z	0.957	0.043
						Sp	0.325	0.675
25	1300	22	air flow	50.0	50.0	Z	0.889	0.111
						Sp	0.324	0.676
26	1350	25	air flow	50.0	50.0	Z	0.826	0.174
						Sp	0.321	0.679
27	1400	48	air flow	50.0	50.0	Z	0.771	0.229
						Sp	0.314	0.686
28	1500	24	air flow	50.0	50.0	Z	0.710	0.290
						Sp	0.301	0.699

Hem = hematite, Sp = spinel solid solution, and Z = zincite solid solution.

powders or from a single-phase spinel solid solution, which was first sintered at a higher temperature (*e.g.*, experiments 3 and 4).

Examples of typical microstructures obtained by sintering a mixture of pure ZnO and Fe_2O_3 powders (experiment 3) and by decomposition of the single-phase spinel solid solution (experiment 4) are presented in Figures 1(a) and (b), respectively. In these figures, the lighter phase is the spinel solid solution and the grey phase is hematite. The shape of the hematite particles is different for the two samples,



(*a*)

(*b*)

Fig. 1—Scanning electron microscopy backscattered electron image of sample with equilibrium between hematite (E) and zinc ferrite (F). (*a*) The sample was prepared from ZnO and Fe_2O_3 and equilibrated at 1000 °C. (*b*) The sample was prepared by decomposition of the zinc ferrite solid solution at 1000 °C.

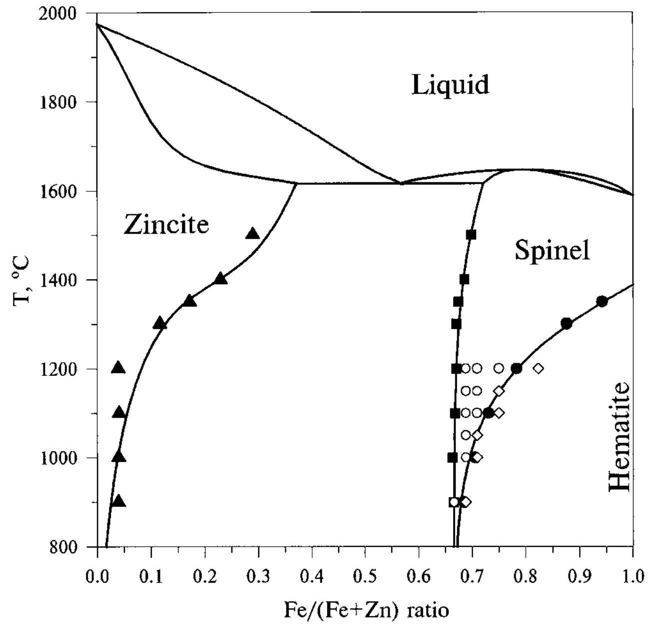


Fig. 2—Zn-Fe-O phase diagram in air. Open circles and open diamonds represent compositions of single-phase spinel and of the spinel + hematite field, respectively, as reported by Tanida and Kitamura.[14] Other points are phase boundaries obtained in the present study. Lines are calculated from optimized model parameters.

reflecting the different paths used to obtain equilibrium. The compositions of the phases measured using EPMA appeared to be identical for samples treated under the same temperature and gas conditions; this verifies the attainment of equilibrium. The comparison of samples quenched in water with those quenched in an air flow showed no sample alteration during quenching (*e.g.*, experiments 4 and 5 and 14 and 15). This fact confirmed that the cooling rate in air was high enough to preserve the phase assemblages that existed at the equilibration temperature.

The measured phase diagram of the Fe-Zn-O system in equilibrium with air is shown in Figure 2. The solid lines are calculated based on the thermodynamic optimization described subsequently. The compositions are expressed in terms of $Zn^{2+}/(Zn^{2+} + Fe^{2+} + Fe^{3+})$ and $(Fe^{2+} + Fe^{3+})/(Zn^{2+} + Fe^{2+} + Fe^{3+})$ mole ratios.

The phase-boundary spinel-Fe_2O_3 obtained in the present study is in good agreement with the quenching experiments performed by Tanida and Kitamura.[14] The compositions of spinel in equilibrium with Fe_2O_3 can also be obtained from the intersection of oxygen isobars in the single-phase spinel field and in the two-phase (spinel + Fe_2O_3) field, which were measured by Yamaguchi and Takei[15] and Benner and Kenworthy.[16] Although the spinel compositions obtained in such a way are not accurate, they are in agreement with the present measurements within experimental error limits.

III. SOLUTION MODELS

A. *Liquid Slag*

A modified quasi-chemical model was used for the molten slag. The model has been described elsewhere.[10–13]

B. *Spinel*

The spinel structure can be derived from the fcc close packing of the oxygen ions. Cations occupy half of the octahedral interstices and one-eighth of the tetrahedral interstices. The ideal stoichiometry is AE_2O_4, where the cations A and E occupy twice as many octahedral sites as tetrahedral ones. A "simple spinel" contains two different cations in the ratio 2:1. If the more abundant of these cations resides on octahedral sites, the spinel is called normal. If it is split evenly between the octahedral and tetrahedral sites, the spinel is called inverse. Generally, both cations in a simple spinel are present on both tetrahedral and octahedral sublattices.

Simple spinels can form extensive solid solutions. Consider a solution formed by the two simple spinels AE_2O_4 and BE_2O_4, where A and B are divalent cations and E is a trivalent cation. Then, a formula unit of the solution is $(A,B,E)[A,B,E]_2O_4$, where the parentheses and brackets correspond to tetrahedral and octahedral sites, respectively. Furthermore, spinels often have an excess of oxygen at higher oxygen pressures, which can be modeled by introducing neutral vacancies (V) on a cation sublattice. Following the earlier studies,[17,18] it was assumed that the vacancies are on the octahedral sublattice, although there is not enough experimental evidence to substantiate this assumption. Usually, the deviation from the ideal stoichiometry toward the metal-rich side is very small and was neglected in the present study, so 1 mole of the solution can be written as

$$(\mathrm{A,B,E})\ [\mathrm{A,B,E,V}]_2\mathrm{O}_4 \qquad [1]$$

The model for spinel was developed within the framework of the CEF. The Gibbs-energy expression in the CEF per formula unit is[19,20]

$$G_m = \sum_i \sum_j Y_i' Y_j'' G_{ij} - TS_c + G_m^E \qquad [2]$$

where Y_i' and Y_j'' represent the site fractions of constituents i and j on the first and second sublattices, respectively, G_{ij} is the Gibbs energy of a pseudocomponent $(i)[j]_2O_4$, and S_c is the configurational entropy,

$$S_c = -R\left(\sum_i Y_i' \ln Y_i' + 2\sum_j Y_j'' \ln Y_j''\right) \qquad [3]$$

and G_m^E is the excess Gibbs energy,

$$G_m^E = \sum_i \sum_j \sum_k Y_i' Y_j' Y_k'' L_{ij:k} + \sum_i \sum_j \sum_k Y_k' Y_i'' Y_j'' L_{k:ij} \qquad [4]$$

where $L_{ij:k}$ and $L_{k:ij}$ are the interaction energies between cations i and j on one sublattice when the other sublattice is occupied only by k.

The application of the CEF to the description of spinels is discussed by Sundman[17] and Barry *et al.*[18] In the present study, a similar treatment of spinels is used; the differences with References 17 and 18 will be outlined subsequently. All pseudocomponents and possible (*i.e.*, charge-neutral) compositions of the solution are shown in Figure 3. The possible compositions of the simple spinel AE_2O_4 lie along the neutral line AE-E[AE] in the plane AE-EE-EA-AA, where the points AE and E[AE] correspond to completely normal and completely inverse spinels, respectively. The possible compositions of a solution between the simple spinels AE_2O_4 and BE_2O_4 form the neutral tetrahedron AE-BE-E[BE]-E[AE]. It should be noted that this is not a plane, as suggested by Barry *et al.*,[18] because three composition variables are required to describe the solution: for example, an overall composition variable, such as the ratio of A to B, plus two internal variables representing the distributions of A and B between octahedral and tetrahedral sites. When vacancies are taken into account, the neutral volume becomes four-dimensional. There is one more neutral apex that lies on the line EE-EV and is marked γ-E_2O_3. This point has the composition $E[E_{5/6}V_{1/6}]_2O_4$ and corresponds to 4/3 moles of the γ modification of the oxide E_2O_3, which has a spinel-related structure.

As can be seen from Eqs. [2] and [4], there are 12 G_{ij} Gibbs energies of pseudocomponents and 30 $L_{ij:k}$ and $L_{k:ij}$ interaction energies, which can be used as parameters to describe the properties of the solution shown in Eq. [1]. Obviously, this is much more than can possibly be obtained from experimental data. Some assumptions have to be made to fix the values of these parameters. It was demonstrated by Barry *et al.*[18] that different sets of parameters can give

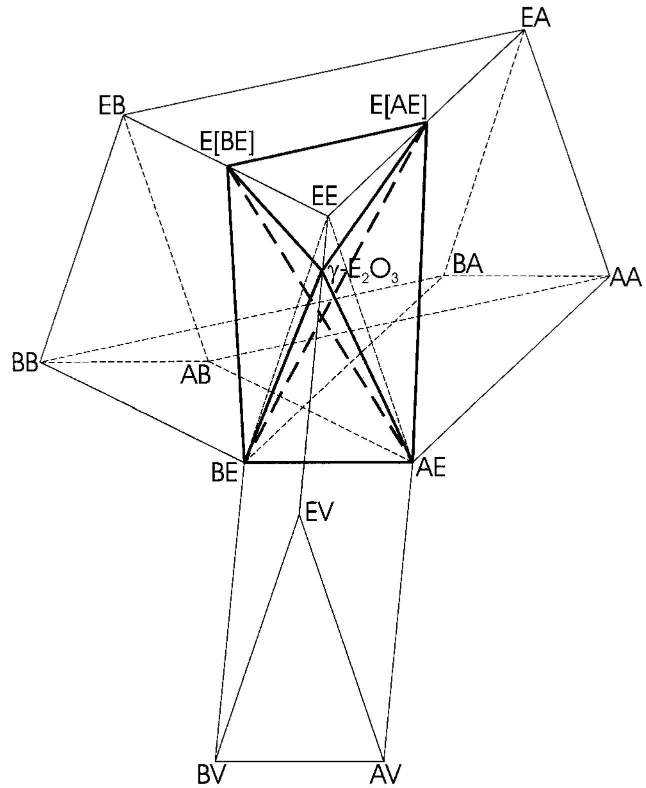


Fig. 3—Pseudocomponents and possible (neutral) composition range of the spinel $(A,B,E)[A,B,E,V]_2O_4$.

equivalent descriptions of the properties of a simple spinel along the neutral line. However, it should be noted that when the properties of a simple spinel are extrapolated into a multicomponent spinel, these different sets of parameters may no longer necessarily give equivalent Gibbs-energy functions, thus resulting in *different models* within the framework of the CEF. For each complex solid solution like a spinel, it is essential to specify a set of model parameters and a *sequence* of their optimization, so that if only a few of them are needed to reproduce the experimental data, the rest of the parameters are kept equal to zero. The following set of model parameters, in the proposed sequence of optimization, is used in the present study for the spinel solution:

$$G_{AE} \qquad [5]$$

$$G_{BE} \qquad [6]$$

$$I_{AE} = G_{EE} + G_{EA} - 2G_{AE} \qquad [7]$$

$$I_{BE} = G_{EE} + G_{EB} - 2G_{BE} \qquad [8]$$

$$\Delta_{AE} = G_{AA} + G_{EE} - G_{EA} - G_{AE} \qquad [9]$$

$$\Delta_{BE} = G_{BB} + G_{EE} - G_{EB} - G_{BE} \qquad [10]$$

$$V_E = G_{EV} - 5/7 G_{EA} \qquad [11]$$

$$\Delta_{EAV} = G_{AA} + G_{EV} - G_{EA} - G_{AV} \qquad [12]$$

$$\Delta_{EBV} = G_{BB} + G_{EV} - G_{EB} - G_{BV} \qquad [13]$$

$$\Delta_{EAB} = G_{AA} + G_{EB} - G_{EA} - G_{AB} \qquad [14]$$

$$\Delta_{EBA} = G_{BB} + G_{EA} - G_{EB} - G_{BA} \qquad [15]$$

$$L_{AB} = L_{AB:A} = L_{AB:B} = L_{AB:E} = L_{AB:V} \quad [16]$$

$$M_{AB} = L_{A:AB}/2 = L_{B:AB}/2 = L_{E:AB}/2 \quad [17]$$

We also adopted the reference state for charged pseudocomponents of spinels that was proposed by Sundman.[17]

$$G_{EA} = G_{AE} \quad [18]$$

where E represents Fe^{3+} and A represents Fe^{2+}. All 12 Gibbs energies of pseudocomponents can be derived from this condition and the first 11 parameters listed previously in Eqs. [5] through [15].

$$G_{AA} = G_{AE} - I_{AE} + \Delta_{AE}$$

$$G_{EE} = G_{AE} + I_{AE}$$

$$G_{EV} = \frac{5}{7} G_{AE} + V_E$$

$$G_{AV} = \frac{5}{7} G_{AE} + V_E - I_{AE} + \Delta_{AE} - \Delta_{EAV}$$

$$G_{EB} = -G_{AE} + 2G_{BE} - I_{AE} + I_{BE}$$

$$G_{BB} = -2G_{AE} + 3G_{BE} - 2I_{AE} + I_{BE} + \Delta_{BE}$$

$$G_{BA} = G_{BE} - I_{AE} + \Delta_{BE} - \Delta_{EBA}$$

$$G_{AB} = -G_{AE} + 2G_{BE} - 2I_{AE} + I_{BE} + \Delta_{AE} - \Delta_{EAB}$$

$$G_{BV} = -\frac{2}{7} G_{AE} + G_{BE} + V_E - I_{AE} + \Delta_{BE} - \Delta_{EBV}$$

The first two parameters, G_{AE} and G_{BE}, are the Gibbs energies of the completely normal, hypothetical simple spinels AE_2O_4 and BE_2O_4, respectively. The parameters I_{AE} and Δ_{AE} are used to describe the degree of inversion and thermodynamic properties of a real simple spinel AE_2O_4. The formation of vacancies resulting in the excess of oxygen in this simple spinel is described by two parameters, V_E and Δ_{EAV}. If the thermodynamic properties of γ-E_2O_3 are available, V_E can be obtained from the equation

$$V_E = 8G_{\gamma-E_2O_3} - 2RT(5 \ln 5 - 6 \ln 6) - 5G_{EE} - 5/7G_{EA} \quad [19]$$

If the properties of γ-E_2O_3 are not known, the parameter V_E can be obtained by fitting experimental data on oxygen nonstoichiometry. One could simply use G_{EV} instead of V_E as a model parameter. However, G_{EV} should normally have some C_P terms. The first approximation for these terms is given by $5/7G_{EA}$, so that V_E is likely to be a linear function of temperature. When one parameter is sufficient for the description of the oxygen nonstoichiometry of the spinel (A,E)[A,E,V]$_2$O$_4$, the second parameter, Δ_{EAV}, can be set equal to zero. For the description of the second spinel (B,E)[B,E,V]$_2$O$_4$ and the solution between the two, there is only one remaining vacancy parameter, Δ_{EBV}. If necessary, two additional excess terms could be used:

$$M_{AV} = L_{A:AV}/2 = L_{B:AV}/2 = L_{E:AV}/2$$

$$M_{BV} = L_{A:BV}/2 = L_{B:BV}/2 = L_{E:BV}/2$$

However, it is believed that in most cases, the parameters of Eqs. [11] through [13] will suffice.

The last four parameters are used for reproducing the properties of the solution between the two simple spinels, including the distribution of cations between octahedral and tetrahedral sites. Obviously, all Δ parameters are Gibbs energies of different exchange reactions of pseudocomponents. Each exchange reaction corresponds to one of the square faces in Figure 3. The choice of which particular exchange reactions are used as model parameters is important if some parameters are kept equal to zero. For example, if $\Delta_{EAB} = 0$, then $\Delta_{AE} = \Delta_{AEB} = G_{EE} + G_{AB} - G_{AE} - G_{EB}$ (Figure 3). On the other hand, if a model parameter $\Delta_{BAE} = G_{AA} + G_{BE} - G_{BA} - G_{AE} = 0$ is used, then $\Delta_{AE} = \Delta_{BEA} = G_{EE} + G_{BA} - G_{EA} - G_{BE}$.

Let a_x and b_x be the total numbers of moles of A and B in one mole of the solution (A,B,E)[A,B,E,V]$_2$O$_4$. The degree of inversion of the solution can be characterized by the internal variables a and b, which are the numbers of moles of A and B on the octahedral sites. The actual values of a and b are obtained by minimizing the Gibbs energy of the solution at a fixed temperature a_x and b_x. We assume that the interaction energy of cations i and j on one sublattice is the same no matter what cation occupies the other sublattice, that is,

$$L_{ij} = L_{ij:r} = L_{ij:s} = \ldots; \quad M_{ij} = L_{r:ij}/2 = L_{s:ij}/2 = \ldots \quad [20]$$

The Gibbs-energy expression (Eq. [2]) can be rearranged in terms of the variables a_x, b_x, a, and b and the model parameters of Eqs. [5] through [17]:

$$\begin{aligned} G_m = {} & a_x G_{AE} + b_x G_{BE} - TS_c + \frac{(1 - a_x - b_x)}{6} \\ & \left(\frac{40}{7} G_{AE} + 5I'_{AE} + V'_E + (a_x - a)(\Delta'_{AE} - \Delta'_{EAV}) + (b_x - b)(\Delta'_{BE} - \Delta'_{EBV}) + aM'_{AV} + bM'_{BV} \right) \\ & + \frac{(a + b)}{2} ((a_x - a)\Delta'_{AE} + (b_x - b)\Delta'_{BE}) + \frac{a}{2} I'_{AE} \\ & + \frac{b}{2} I'_{BE} - \frac{(a_x - a)b}{2} \Delta'_{EAB} - \frac{(b_x - b)a}{2} \Delta'_{EBA} \\ & + \frac{ab}{2} M_{AB} + (a_x - a)(b_x - b)L_{AB} \end{aligned} \quad [21]$$

where

$$\begin{aligned} \Delta'_{AE} &= \Delta_{AE} + 2L_{AE} + M_{AE} \\ I'_{AE} &= I_{AE} + M_{AE} \\ V'_E &= V_E - 5M_{AE} + 5/3M_{EV} \\ \Delta'_{EAB} &= \Delta_{EAB} - M_{BE} + M_{AE} \\ \Delta'_{EBA} &= \Delta_{EBA} - M_{AE} + M_{BE} \\ \Delta'_{EAV} &= \Delta_{EAV} - 4L_{AE} + M_{AE} - M_{EV}/3 \\ M'_{AV} &= M_{AV} - 2/3M_{EV} + 2M_{AE} \\ & etc. \end{aligned} \quad [22]$$

The condition of electroneutrality is taken into account when the site fractions (Y_i) are expressed in terms of the independent variables a_x, b_x, a, and b. Therefore, Eq. [21] gives the Gibbs energy for a neutral volume in Figure 3. Obviously, when the conditions of Eq. [20] are used and only the neutral volume is considered, all the interaction energies but four (L_{AB}, M_{AB}, M_{AV}, and M_{BV}) are no longer independent parameters, since only the combinations of Eq. [22] can be obtained from experimental data. Therefore, without any loss of generality, those interaction energies can be set equal to zero. It should be noted that if the conditions of Eq. [20] are not used, the Gibbs-energy expression becomes unreasonably complicated; even when vacancies are not taken into account, nine additional terms appear, such as $(a_x)^2a$, a_xa^2, a^3, *etc.* With vacancies, the number of additional terms doubles. In our experience, such an excessive number of adjustable parameters can result in unrealistic distributions of cations between sublattices. In any case, for any real spinel solution, there are never enough experimental data to obtain all these parameters. Therefore, it is recommended that the conditions of Eq. [20] always be obeyed. For example, this implies the automatic addition of the parameter $L_{ij:k} = L_{ij}$ if a new component k is added to a spinel solution and the interaction energy L_{ij} has already been fixed.

If the oxygen nonstoichiometry is ignored, then $a_x + b_x = 1$. When, also, $\Delta'_{EAB} = \Delta'_{EBA} = 0$ and $\Delta'_{AE} = \Delta'_{BE} = \Delta$, Eq. [21] can be written as

$$G_m = a_xG_{AE} + b_xG_{BE} - TS_c + a\left(\frac{\Delta + I'_{AE}}{2} - \frac{\Delta}{2}(a+b)\right) + b\left(\frac{\Delta + I'_{BE}}{2} - \frac{\Delta}{2}(a+b)\right) + \frac{ab}{2}M_{AB} + (a_x - a)(b_x - b)L_{AB} \quad [23]$$

If we set $M_{AB} = L_{AB} = 0$ and introduce the notation $\beta = -\Delta/2$, $\alpha_{A-E} = (\Delta + I'_{AE})/2$, and $\alpha_{B-E} = (\Delta + I'_{BE})/2$, Eq. [23] reduces to the model proposed by O'Neill and Navrotsky,[21,22] which has been successfully used for many solutions of two simple spinels. On the other hand, Barry *et al.*[18] mentioned that their spinel model based on the CEF could be reduced to the model by O'Neill and Navrotsky for a simple spinel but not for a solution of two simple spinels. This reinforces the importance of obeying the conditions of Eq. [20] and selecting appropriate model parameters within the framework of the CEF.

For a perfectly normal solution of two normal simple spinels, a and b are equal to zero, and the Gibbs-energy expression is given by

$$G_m = a_xG_{AE} + b_xG_{BE} - TS_c + a_xb_xL_{AB} \quad [24]$$

There is only one parameter, L_{AB}, left to describe the properties of this solution. Similarly, for a strictly inverse solution of two inverse simple spinels, $a_x = a$ and $b_x = b$, and the Gibbs energy can be written as

$$G_m = a_x(G_{AE} + I'_{AE}/2) + b_x(G_{BE} + I'_{BE}/2) - TS_c + a_xb_xM_{AB}/2 \quad [25]$$

If the radii of cations A and B are nearly equal, the solutions are likely to be ideal, so that $L_{AB} = 0$ and $M_{AB} = 0$. However, when the difference in ionic radii is significant, the interaction energies L_{AB} and M_{AB} become very important.

The effect of size mismatch was taken into account by O'Neill and Navrotsky[21,22] by using a regular-solution term, a_xb_xW. However, they neglected the effect of this term on the distribution of cations between the octahedral and tetrahedral sites. This is probably an oversimplification. On the other hand, the Gibbs-energy expression of Eq. [23] takes this into account and can be regarded as a natural extension of the O'Neill and Navrotsky treatment of size mismatch. If a solution is nearly normal or nearly inverse, then one should use either the parameter L_{AB} or M_{AB}, respectively. For the intermediate case, if it is not possible to obtain both parameters from experimental data, one can use the approximation $L_{AB} = M_{AB}$, as suggested by Barry *et al.*,[18] which is based on the fact that there are twice as many cations on the octahedral sites as on the tetrahedral sites. However, it should be realized that this is a very rough approximation, because the coordination numbers and ionic radii are different for cations on these two sublattices. When the ionic radii of A and B are nearly equal, it is better to set L_{AB} and M_{AB} equal to zero and to use the parameters Δ_{EAB} and Δ_{EBA} to reproduce the properties of a solution.

The set of parameters proposed here in Eqs. [5] through [17] is not unique. Other choices can be made within the framework of the CEF, which would result in different Gibbs-energy expressions in terms of independent variables. The present choice is corroborated by two considerations. First, the assumptions $\Delta_{EAB} = \Delta_{EBA} = \Delta_{EAV} = 0$ gave very good results for the binary spinel solution in the Fe-Zn-O system. In fact, only parameters corresponding to the two limiting simple spinels were sufficient to reproduce the properties of this solution. Second, the model by O'Neill and Navrotsky,[22] which has proved successful in describing the major features of binary spinel solutions, can be obtained as a limiting case of the model presented here. Of course, with the parameters in Eqs. [5] through [17], it is possible to reproduce the properties of spinel solutions much more precisely than with the limiting-case model. The appropriate set and sequence of model parameters for solutions $(A,E,F)[A,E,F,V]_2O_4$ and $(A,B,E,F)[A,B,E,F,V]_2O_4$, where A and B are divalent cations and E and F are trivalent cations, have also been worked out and will be described elsewhere. This is sufficient to model spinel solutions with any number of components.

The spinel solid solution in the Fe-Zn-O system exhibits magnetic ordering. Magnetite is ferromagnetic below its Curie temperature of 848 K, and zinc ferrite is antiferromagnetic below its Neel temperature of 9.5 K. The phenomenological approach proposed by Hillert and Jarl[23] was used to describe the magnetic contribution to the thermodynamic functions. The magnetic contribution to the Gibbs energy is given by

$$G^{magn} = f(\tau)RT\ln(\beta + 1) \quad [26]$$

where

$$\tau = T/T_c$$

$$f(\tau) = 1 - \left(\frac{79\tau^{-1}}{140p} + \frac{474}{497}\left(\frac{1}{p} - 1\right)\left(\frac{\tau^3}{6} + \frac{\tau^9}{135} + \frac{\tau^{15}}{600}\right)\right)/A, \quad \text{for } \tau \leq 1$$

$$f(\tau) = -\left(\frac{\tau^{-5}}{10} + \frac{\tau^{-15}}{315} + \frac{\tau^{-25}}{1500}\right)/A, \quad \text{for } \tau > 1$$

$$A = \frac{518}{1125} + \frac{11{,}692}{15{,}975}\left(\frac{1}{p} - 1\right)$$

where

T_c = the critical temperature for magnetic ordering,
β = the average magnetic moment per atom, and
p = a structure-dependent constant.

In general, T_c and β are functions of the overall composition and distribution of cations between the octahedral and tetrahedral sublattices.

When the CEF is used to describe a solution,[24] T_c and β are given by equations similar to the Gibbs-energy expression:

$$\beta = \sum_i \sum_j Y_i' Y_j'' \beta_{ij} + \sum_i \sum_j \sum_k Y_i' Y_j' Y_k'' \beta_{ij:k} + \sum_i \sum_j \sum_k Y_k' Y_i'' Y_j'' \beta_{k:ij} \qquad [27]$$

where β_{ij}, $\beta_{ij:k}$, and $\beta_{k:ij}$ are parameters to describe the concentration dependence of β. A similar equation is written for T_c. These equations do not have any physical meaning, because T_c and β are not energies. Using similar equations and similar types of parameters simply makes the programming of general thermodynamic software easier. However, as will be shown subsequently, this results in the need to introduce many interaction parameters just to ensure that Eq. [27] gives the most simple linear composition dependence of β.

Experimental information on the concentration dependence of T_c and β is very limited. It is necessary to assign values to the parameters β_{ij}, $\beta_{ij:k}$, and $\beta_{k:ij}$ based on limited experimental data, in such a way that Eq. [27] gives a reasonable approximation for the β value of the solution. As a first approximation, we can use the values of β at five apexes of the neutral volume in Figure 3 and assume that β changes linearly along the edges of the neutral volume. To obtain such a simple model, it is necessary to introduce the following parameters:

$$\begin{aligned}
\beta_{\mathrm{EA}} &= 2\beta^i_{\mathrm{AE}} \\
\beta_{\mathrm{EB}} &= 2\beta^i_{\mathrm{BE}} \\
\beta_{\mathrm{EV}} &= 8\beta_{\mathrm{E_2O_3}} \\
\beta_{\mathrm{AE:A}} &= \beta_{\mathrm{AE:E}} = \beta_{\mathrm{AE}}/2 + \beta^i_{\mathrm{AE}} \\
\beta_{\mathrm{BE:B}} &= \beta_{\mathrm{BE:E}} = \beta_{\mathrm{BE}}/2 + \beta^i_{\mathrm{BE}} \\
\beta_{\mathrm{AE:V}} &= 4/3\beta_{\mathrm{E_2O_3}} + \beta_{\mathrm{AE}}/6 \\
\beta_{\mathrm{BE:V}} &= 4/3\beta_{\mathrm{E_2O_3}} + \beta_{\mathrm{BE}}/6 \\
\beta_{\mathrm{A:E}V} &= 8\beta_{\mathrm{E_2O_3}} - 2\beta_{\mathrm{AE}} - 6\beta^i_{\mathrm{AE}} \\
\beta_{\mathrm{B:E}V} &= 8\beta_{\mathrm{E_2O_3}} - 2\beta_{\mathrm{BE}} - 6\beta^i_{\mathrm{BE}} \\
\beta_{\mathrm{AE:B}} &= \beta_{\mathrm{AE}}/2 + \beta^i_{\mathrm{BE}} \\
\beta_{\mathrm{BE:A}} &= \beta_{\mathrm{BE}}/2 + \beta^i_{\mathrm{AE}} \\
\beta_{\mathrm{A:BE}} &= -\beta_{\mathrm{B:AE}} = 2(\beta^i_{\mathrm{BE}} - \beta^i_{\mathrm{AE}})
\end{aligned} \qquad [28]$$

where β_{AE} and β^i_{AE} are the values of β for the normal and inverse simple spinel AE_2O_4, respectively, and $\beta_{\mathrm{E_2O_3}}$ is the magnetic moment for γ-E_2O_3. The other parameters in Eq. [27] must be set equal to zero. Of course, if $\beta_{\mathrm{AE}} = \beta^i_{\mathrm{AE}}$, the magnetic moment will be constant along the neutral line representing the simple spinel AE_2O_4. In rather rare cases when it is desirable to have a quadratic dependence of β as a function of the overall composition, one can use the single parameter β_{AB}, which is defined as follows:

$$\begin{aligned}
\beta_{\mathrm{AB:E}} &= \beta_{\mathrm{E:AB}}/2 = \beta_{\mathrm{AB}} \\
\beta_{\mathrm{AE:B}} &= \beta_{\mathrm{AE}}/2 + \beta^i_{\mathrm{BE}} + \beta_{\mathrm{AB}} \\
\beta_{\mathrm{BE:A}} &= \beta_{\mathrm{BE}}/2 + \beta^i_{\mathrm{AE}} + \beta_{\mathrm{AB}} \\
\beta_{\mathrm{A:BE}} &= 2(\beta^i_{\mathrm{BE}} - \beta^i_{\mathrm{AE}} + \beta_{\mathrm{AB}}) \\
\beta_{\mathrm{B:AE}} &= 2(\beta^i_{\mathrm{AE}} - \beta^i_{\mathrm{BE}} + \beta_{\mathrm{AB}})
\end{aligned} \qquad [29]$$

where all the other parameters are given by Eq. [28]. Similar parameters should be used to describe the composition dependence of T_c.

When the effect of pressure is considered, the use of the Murnaghan or the Birch–Murnaghan equation with the CEF has been suggested.[24] The compositional dependence of the parameters of these equations is also expressed in the form of Eq. [27], and, hence, Eqs. [28] or [29] can be used to obtain the simplest first approximations.

C. *Wustite and Zincite*

The Gibbs energies of wustite and zincite were represented by the polynomial model with the Kohler-like[25,26] extension of binary and ternary terms into multicomponent systems:

$$G_m = \sum_i X_i G_i^\circ + RT \sum_i X_i \ln X_i + \sum_i \sum_j X_i X_j \left(\frac{X_i}{X_i + X_j}\right)^m \left(\frac{X_j}{X_i + X_j}\right)^n L_{ij}^{mn} + \sum_i \sum_j \sum_k X_i X_j X_k \left(\frac{X_i}{X_i + X_j + X_k}\right)^l \left(\frac{X_j}{X_i + X_j + X_k}\right)^m \left(\frac{X_k}{X_i + X_j + X_k}\right)^n L_{ijk}^{lmn} \qquad [30]$$

where X_i and G_i° are the mole fraction and the Gibbs energy of component i; L_{ij}^{mn} and L_{ijk}^{lmn} are interaction parameters, which can be temperature dependent; and the powers l, m, and n are ≥ 0. The components of wustite and zincite in the Fe-Zn-O system are FeO, ZnO, and $FeO_{1.5}$.

IV. CRITICAL EVALUATION AND OPTIMIZATION

A. *The Fe-O System*

The thermodynamic properties and phase diagram of the Fe-O system were reviewed by Spencer and Kubaschewski[27] and optimized by Sundman[17] using the CEF for the solid phases and an ionic two-sublattice model for the liquid. In the present study, the quasi-chemical model,[10–13] the polynomial model, and the CEF are used for the oxide melt, wustite, and magnetite, respectively. The thermodynamic properties of the FeO-Fe_2O_3 liquid and wustite were optimized earlier,[28] and the parameters of the models are given

Table II. The Thermodynamic Properties of the Spinel, Wustite, and Zincite Solid Solutions

Parameters of the Model	Value, J/mol
	Spinel: $(Fe^{2+}, Fe^{3+}, Zn)[Fe^{2+}, Fe^{3+}, Zn, V]_2O_4$
$G_{Fe^{2+}Fe^{3+}}$	$-1{,}141{,}701 + 1014.54T - 0.00814919T^2$ $-174.832T \ln(T) + 1{,}445{,}276/T$
$G_{ZnFe^{3+}}$	$-1{,}175{,}354 + 1615.28T - 5186.49T^{0.5}$ $-7359.34 \ln(T) - 237.559T \ln(T)$
$I_{Fe^{2+}Fe^{3+}}$	$-31{,}229 + 22.063T$
$I_{ZnFe^{3+}}$	51,800
$V_{Fe^{3+}}$	$29{,}932 + 28.547T$
$\Delta_{Fe^{2+}Fe^{3+}}$	15,781
$\Delta_{ZnFe^{3+}}$	40,000
$\Delta_{Fe^{3+}ZnV}$	−36,000
	Magnetic properties
$\beta_{Fe^{2+}Fe^{3+}} = \beta^i_{Fe^{2+}Fe^{3+}}$	44.54
$T_{Fe^{2+}Fe^{3+}} = T^i_{Fe^{2+}Fe^{3+}}$	848
$\beta_{ZnFe^{3+}} = \beta^i_{ZnFe^{3+}}$	-3×3.987*
$T_{ZnFe^{3+}} = T^i_{ZnFe^{3+}}$	-3×9.5*
p	0.28
	Wustite: $FeO\text{-}FeO_{1.5}\text{-}ZnO$
$G^{o(Wustite)}_{ZnO} - G^{o(Zincite)}_{ZnO}$	12,552
$L^{00}_{FeO\text{-}FeO_{1.5}}$	$8577.2 - 9.665T$
$L^{02}_{FeO\text{-}FeO_{1.5}}$	10,460
$L^{00}_{ZnO\text{-}FeO_{1.5}}$	12,552
$L^{002}_{FeO\text{-}ZnO\text{-}FeO_{1.5}}$	−376,560
	Zincite: $ZnO\text{-}FeO\text{-}FeO_{1.5}$
$G^{o(Zincite)}_{FeO}\text{-}G^{o(Wustite)}_{FeO}$	16,736
$G^{o(Zincite)}_{FeO_{1.5}}\text{-}G^{o(Hematite)}_{FeO_{1.5}}$	3766
$L^{00}_{FeO\text{-}ZnO}$	−6276
$L^{00}_{ZnO\text{-}FeO_{1.5}}$	−15,071
$L^{10}_{ZnO\text{-}FeO_{1.5}}$	38,589
	Liquid Oxide: $FeO\text{-}FeO_{1.5}\text{-}ZnO$
$\omega^{00}_{FeO\text{-}ZnO}$**	15,895
$\omega^{01}_{FeO\text{-}FeO_{1.5}}$	$13{,}694 - 16.736T$
$\omega^{07}_{FeO\text{-}FeO_{1.5}}$	$129{,}704 - 60.250T$

*Negative T_c and β should be divided by −3 before calculation of G^{magn} from Eq. [26].

**The quasi-chemical ω parameters are defined in Refs. 10 through 13. Gibbs energies of pure components of wustite, zincite, and liquid oxide are taken from the F*A*C*T database.[1]

in Table II. The properties of magnetite and hematite were optimized in the course of the present study. The thermodynamic functions of all stoichiometric phases and vapor species that are not given in this article were taken from the F*A*C*T database.[1]

Hematite (Fe_2O_3) has a trigonal structure based on the hcp oxygen-packing scheme. The space group is $R\bar{3}c$.[29] Hematite is antiferromagnetic below a Neel temperature of 956 K. The magnetic contribution to the thermodynamic functions is given by Eq. [26]. The nonstoichiometry of hematite is negligible,[17] and it was assumed to be a stoichiometric compound. The thermodynamic properties, including magnetic properties, were taken from the optimization of Sundman.[17] However, the enthalpy of formation and entropy at 298.15 K were slightly changed from the values used by Sundman, in order to make the properties of Fe_2O_3 consistent with the optimized thermodynamic functions of the oxide liquid.[28] This made them a little closer to the JANAF values.[30] The optimized values for hematite are

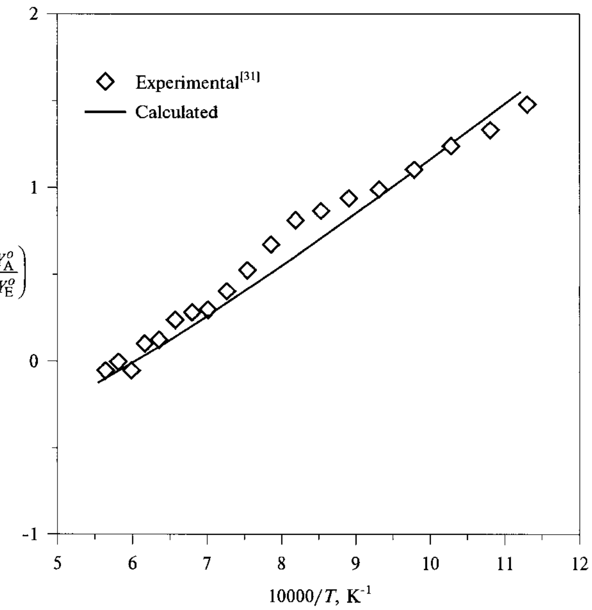


Fig. 4—Degree of order in Fe_3O_4.

$$\Delta H_f^\circ(298.15) = -825{,}787 \text{ J/mol}$$

$$S^\circ(298.15) = 87.7 \text{ J/mol/K}$$

$$C_p = 137.009 - 2{,}907{,}640\ T^{-2} \text{ J/mol/K}$$

The model parameters for magnetite are given in Table II. The Curie temperature and the average magnetic moment were taken from Sundman.[17] A good description of the heat capacity and heat content was obtained, similar to that reported by Sundman.[17]

The most reliable data on the degree of the inversion of magnetite were obtained by Wu and Mason[31] from the temperature variation of the thermopower. These data were fitted by Sundman,[17] assuming $\Delta_{AE} = 0$. This resulted in a temperature dependence of his parameter $B(Fe_3O_4)$, which is unusually high as compared to many other spinels.[21,22] Therefore, we used two parameters, I_{AE} and Δ_{AE}, to describe the data.[31] Although the temperature dependence of the parameter I_{AE} is much smaller when Δ_{AE} is not equal to zero, it is still necessary to have a nonzero temperature-dependent term to reproduce the data. As was pointed out by O'Neill and Navrotsky,[22] Mössbauer spectroscopy was unable to resolve Fe^{2+} and Fe^{3+} on the octahedral sites. This implies that the rate of electron hopping is faster than the precession of the iron nucleus ($\sim 10^{-7}$ seconds), which raises the question whether Fe^{2+} and Fe^{3+} can be considered as distinguishable species in a thermodynamic sense. These ions have to be assumed to be distinguishable if the CEF is used to model the spinel. It is interesting to mention that the obtained temperature dependence of the parameter I_{AE} is almost equal to the value that would be necessary to compensate the entropy originating from the random mixing of distinguishable Fe^{2+} and Fe^{3+} ions. The calculated degree of inversion is compared to experimental data[31] in Figure 4.

The excess of oxygen in magnetite is described by the temperature-dependent parameter V_E. The second parameter, Δ_{EAV}, is set equal to zero. The calculated iron deficiency in magnetite and the experimental measurements are shown in

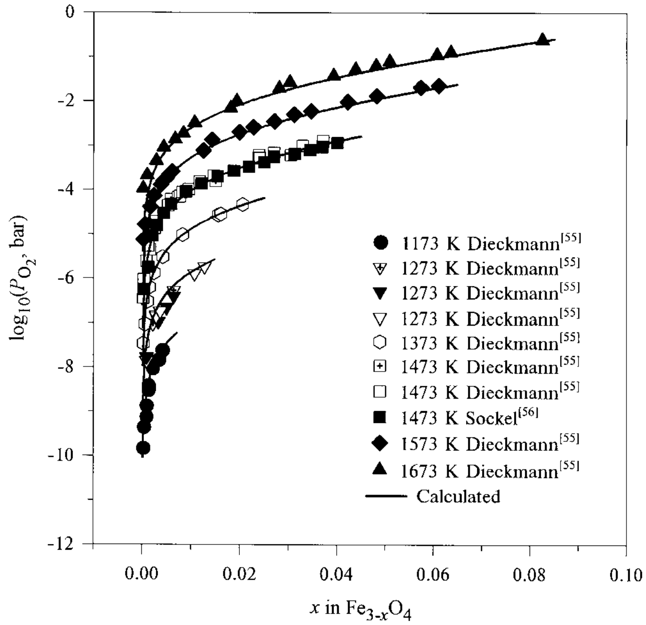


Fig. 5—Experimental and calculated oxygen partial pressure over single-phase magnetite as a function of composition.

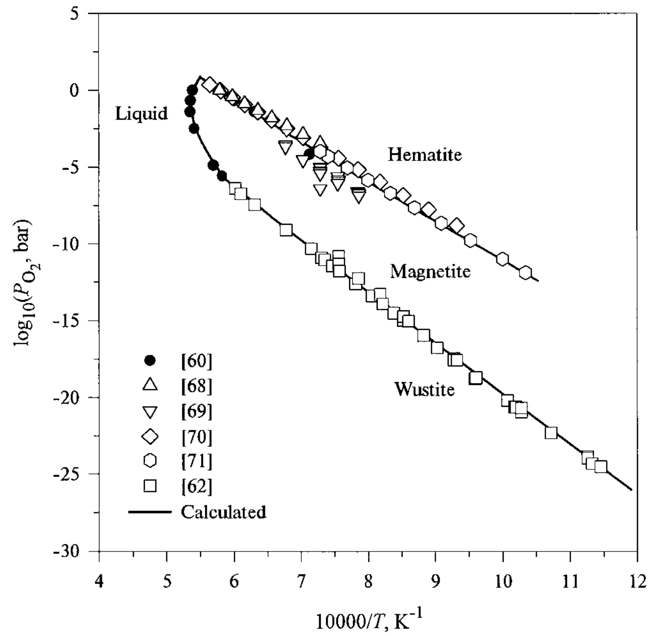


Fig. 7—Oxygen partial pressure for two-phase equilibria with magnetite in the Fe-O system.

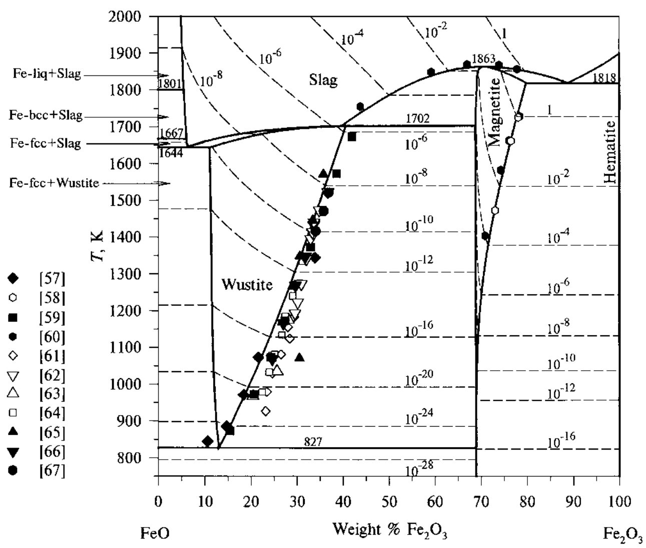


Fig. 6—FeO-Fe_2O_3 phase diagram: selected experimental points and calculated lines and invariant temperatures. Dashed lines are calculated oxygen isobars (bar).

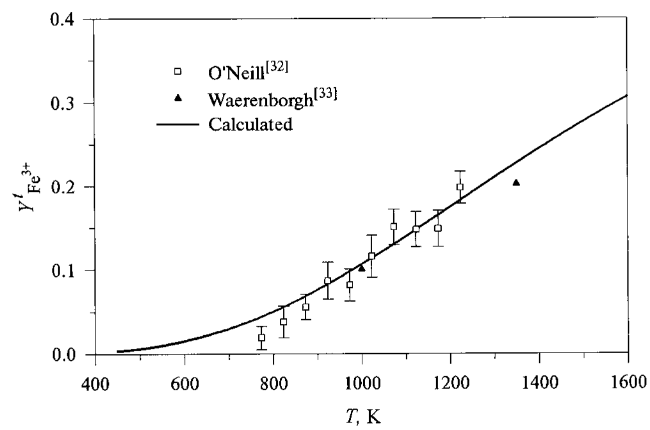


Fig. 8—Experimental and calculated degree of inversion of $ZnFe_2O_4$ spinel. Error bars are ± 1 standard deviation.

Figure 5. The FeO-Fe_2O_3 phase diagram is presented in Figure 6. As can be seen from Figures 6 and 7, the calculated phase boundaries among hematite, magnetite, wustite, and liquid oxide are in good agreement with the experiments. Experimental data on the oxygen potentials in the single-phase wustite and in the liquid slag, as well as the phase boundaries among wustite, slag, and iron, are also well represented by the optimized equations as described by Wu.[28]

B. *The Fe-Zn-O System*

In the spinel solid solution, there is a complete range of solid solubility between magnetite and zinc ferrite. The distribution of cations between octahedral and tetrahedral sites in quenched samples of zinc ferrite was studied by powder X-ray diffraction, using the Rietveld method of structural refinement.[32,33] O'Neill[32] obtained similar descriptions of his data with three different combinations of model parameters. We used the description based on the value of $\Delta_{ZnFe^{3+}} = 40$ kJ, which was suggested by O'Neill and Navrotsky[22] for 2:3 spinels. The calculated site fraction of Fe^{3+} on tetrahedral sites is compared with experimental data in Figure 8. O'Neill pointed out that the rate of reordering was probably too fast to quench the cation distribution above 950 °C. Therefore, his data at higher temperatures are not shown in the figure.

The low-temperature heat capacity of zinc ferrite was measured calorimetrically over the range from 5 to 350 K[34] and 51 to 298 K.[35] The lambda transition corresponding to antiferromagnetic ordering was detected at 9.5 K. The magnetic contribution to the Gibbs energy of zinc ferrite was described by Eq. [26], where the values of $T_c = 9.5$ K

Table III. Integral Thermodynamic Functions of Zinc Ferrite

Thermodynamic Function	T, K	Calculated (J/mol)	Experimental (J/mol)	Reference	Experimental Method
S^o	298.15	150.7	150.7	34	low temperature heat capacity
ΔH_f^{ox}	970	−17,220	−11,170	38	solution calorimetry in $9PbO{\cdot}3CdO{\cdot}4B_2O_3$
ΔH_f^{ox}	413	−8,820	−29,290	39 cited from 42	solution calorimetry in $H_2SO_4 + H_3PO_4$
ΔH_f^{ox}	673	−11,350	4,900	39 cited from 42	calorimetry, reduction of $ZnFe_2O_4$ and Fe_2O_3 by H_2
ΔH_r	698	90,980	72,300	40	calorimetry of the reaction $ZnFe_2O_4 + 3H_2$ (g) = ZnO + 2α-Fe + $3H_2O$ (g)
ΔG_f^{ox}	1367	−23,480	−27,150	41	equilibria with HCl (g)
ΔG_f^{ox}	1484	−24,460	−27,150	41	equilibria with HCl (g)
ΔG_f^{ox}	1073	−21,730	−20,590	42	$P(O_2)$ for spinel + zincite and spinel + wustite equilibria
ΔG_f^{ox}	1373	−23,530	−20,970	42	$P(O_2)$ for spinel + zincite and spinel + wustite equilibria

ΔH_f^{ox} and ΔG_f^{ox} are the enthalpy and Gibbs energy of formation of zinc ferrite from ZnO and Fe_2O_3, respectively.

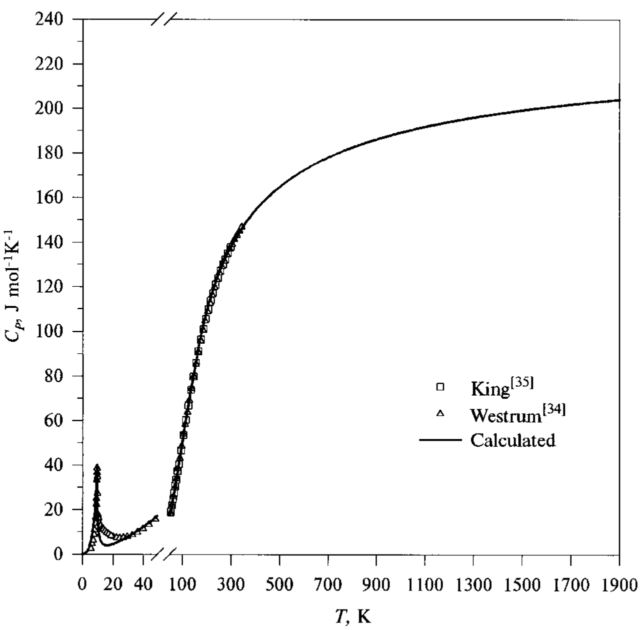


Fig. 9—Experimental and calculated C_P of the normal $ZnFe_2O_4$ spinel.

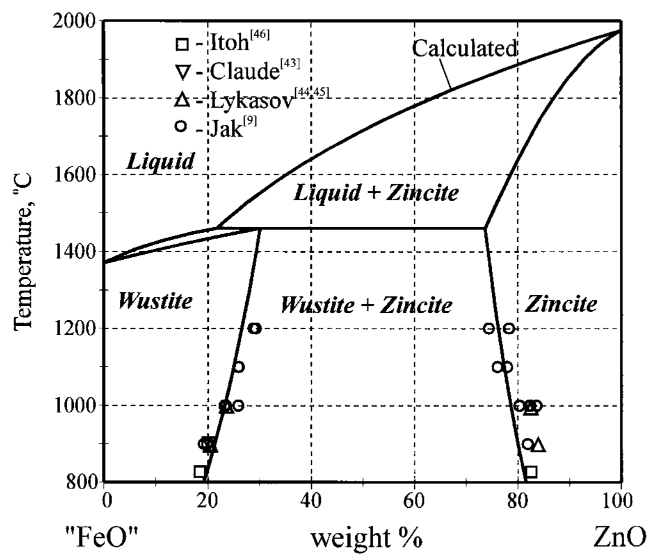


Fig. 10—Phase diagram "FeO"-ZnO at metallic iron saturation (all lines are calculated from the model).

and $\beta = 3.987$ were obtained by fitting experimental data on the heat capacity. The structure-dependent parameter p was set equal to 0.28 for the fcc spinel structure.[17] As can be seen from Figure 9, the calculated C_P value is slightly lower than the experimental data in the vicinity of the magnetic transition, because of limitations of the magnetic model. However, the resulting error in the Gibbs energy is negligible. The standard entropy at 298.15 K, calculated by Westrum and Grimes[34] from their heat-capacity data, was accepted for the present optimization (Table III).

The high-temperature heat capacity of zinc ferrite was reported in References 36 and 37. These data were averaged and joined smoothly with the low-temperature C_P.

The enthalpy of zinc ferrite was studied by solution calorimetry in molten oxide solvents[38] and in a mixture of sulphuric and phosphoric acids.[39] The enthalpy values can also be derived from the measured heats of reduction of zinc ferrite by hydrogen.[39,40] As can be seen from Table III, the reported enthalpies are contradictory. Gilbert and Kitchener[41] calculated the Gibbs energy of $ZnFe_2O_4$ from that of $ZnAl_2O_4$ and the study of high-temperature equilibria of these compounds with hydrogen chloride. Tretyakov[42] derived similar values of the Gibbs energy from his electromotive force (EMF) measurements of the oxygen partial pressures over spinel in equilibrium with zincite and wustite. In the present assessment, more weight was given to the Gibbs-energy measurements. The thermodynamic properties of real zinc ferrite were reproduced by optimizing the Gibbs-energy expression $G_{ZnFe^{3+}}$ for the hypothetical normal zinc ferrite (Table II). The calculated and experimental Gibbs energies and enthalpies are compared in Table III.

The EPMA measurements of the compositions of zincite and wustite in equilibrium with iron were reported by Jak and Hayes.[9] As can be seen from Figure 10, these data are in good agreement with the results of other studies.[43–46] The lines in the figure are calculated from the optimized model parameters. All measured iron in wustite and zincite was assumed to be "FeO;" therefore, Figure 10 shows the

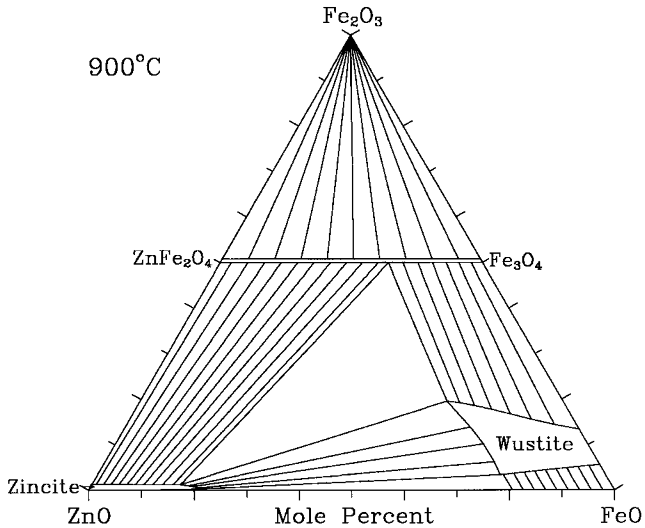
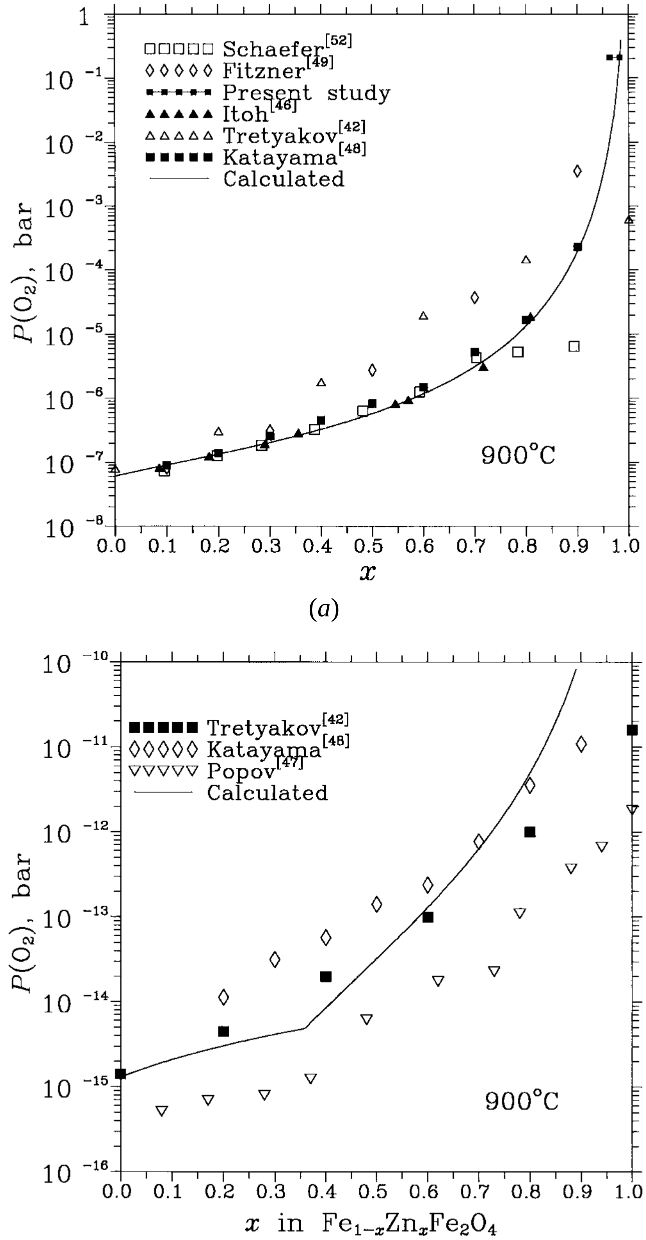


Fig. 11—ZnO-FeO-Fe_2O_3 phase diagram at 900 °C.



(*a*)

(*b*)

Fig. 12—(*a*) Equilibrium oxygen pressure for the spinel + Fe_2O_3 two-phase region at $Fe_{1-x}Zn_xFe_2O_4$ + Fe_2O_3 overall composition. (*b*) Equilibrium oxygen pressure for the spinel + zincite and spinel + wustite two-phase regions at 900 °C.

projection of the phase equilibria in the Fe-Zn-O system onto the FeO-ZnO join through the oxygen corner of the composition triangle.

The calculated ZnO-FeO-Fe_2O_3 phase diagram at 900 °C is shown in Figure 11. The phase equilibria among hematite and three solid solutions (spinel, wustite, and zincite) are of primary importance in this system. The studies of equilibrium oxygen partial pressure over single-phase, two-phase, and three-phase fields are abundant.[42–53] Most of the data were obtained by EMF measurements with CaO-stabilized or yttria-stabilized ZrO_2 solid electrolytes.[42,44–46,48–53]

It should be noted that the phase compositions reported in the EMF studies can be affected by zinc oxide losses due to evaporation of ZnO at high temperatures during both the preparation of electrodes and EMF measurements. Furthermore, when Fe_2O_3 or wustite were sintered with the spinel solid solution to make an electrode, they could, respectively, reduce or oxidize to form Fe_3O_4, depending on the sintering conditions. The latter would then dissolve in the spinel, changing its composition. Similarly, the preparation of the (spinel + ZnO) electrode should result in dissolution of some amount of FeO in zincite with a corresponding change in the spinel composition.

As can be seen from Figures 12(a) and (b), experimental data on the equilibrium oxygen pressures are widely scattered. Similar data were reported at 800 °C, 1000 °C, and 1100 °C.[42,48] The equilibrium oxygen partial pressure on the reduced side of the spinel solid solution was also obtained from the measurements of equilibrium constants for the reduction of $ZnFe_2O_4$ by hydrogen at 600 °C, 700 °C, and 900 °C.[47] In this case, the composition of spinel was derived from the crystal lattice constants. This could result in substantial error. Therefore, in the current optimization, more weight was assigned to experimental data on the phase equilibria in air and at iron saturation (Figures 2 and 10).

The EMF measurements of the equilibrium oxygen partial pressures over single-phase wustite and over two-phase regions (wustite + zincite) and (wustite + iron) were performed by Lykasov *et al.*[45,50] at 850 °C to 1000 °C. They also obtained the oxygen potentials for the univariant equilibria (Fe + wustite + zincite) and (spinel + wustite + zincite).[44] Claude *et al.*[43] studied wustite in equilibrium with iron and zincite at 900 °C. Itoh *et al.*[46,53] reported the oxygen partial pressures for all the two- and three-phase fields shown in Figure 11 at 827 °C. Lykasov *et al.*[44,45,50] suggested that their powder X-ray diffraction results indicated the phase separation of wustite into two phases. However, this was not observed in any other studies.

All these data were used to optimize parameters of the models for wustite, zincite, and spinel solid solutions. The model parameters obtained for wustite and zincite are given in Table II. No additional parameters were necessary for the spinel; *i.e.*, the thermodynamic functions of spinel extrapolated from pure magnetite and zinc ferrite provide a good

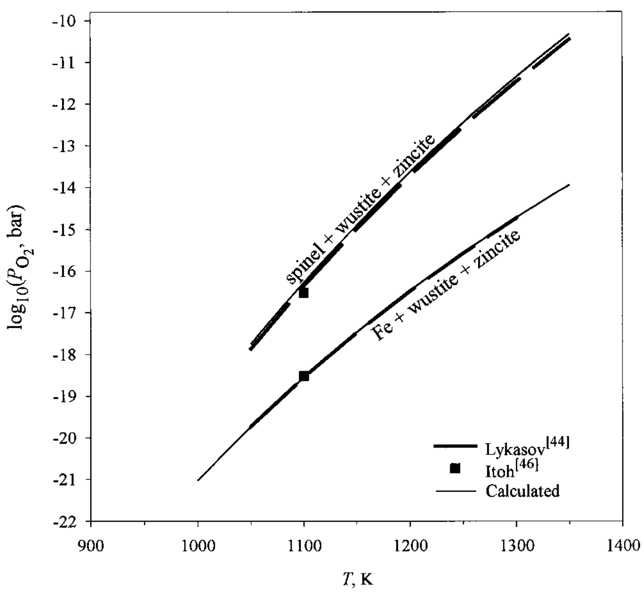


Fig. 13—Oxygen potential for three-phase equilibria in the Fe-Zn-O system.

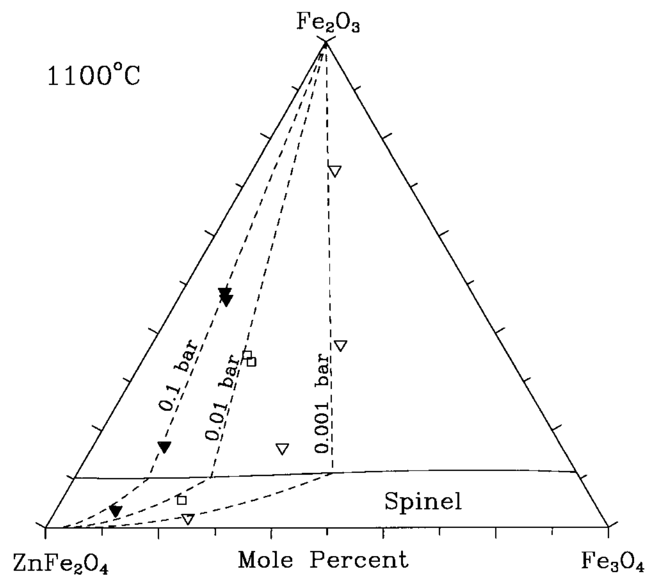


Fig. 14—Phase equilibria between spinel and Fe_2O_3 at 1100 °C. Phase boundary and oxygen isobars are calculated. Points are experimental data.[16]

approximation of the thermodynamic properties of the solution. All experimental data are reproduced within experimental error limits (*e.g.*, Figures 2, 10, 12, and 13).

The phase-equilibrium data obtained in the experimental part of the present study are also well represented by the model parameters (Figure 2). It was not possible to describe the phase boundaries of zincite without accepting the presence of some Fe^{3+} in this solution. Although the Fe^{3+} concentration in zincite at 900 °C is small, as can be seen from Figure 11, it increases at about 1300 °C, causing an inflection of the zincite phase boundary in equilibrium with spinel (Figure 2).

The parameter $\Delta_{Fe^{3+}ZnV}$ of the model for zinc ferrite was selected to provide about the same solubility of Fe_2O_3 in $ZnFe_2O_4$ as in Fe_3O_4. The solubility of Fe_2O_3 in the spinel solid solution can be estimated from the oxygen isobars in the Fe_3O_4-$ZnFe_2O_4$-Fe_2O_3 system reported in References 15 and 16 from 1050 °C to 1400 °C. In these studies, initial mixtures of ZnO and Fe_2O_3 were equilibrated at a fixed temperature and oxygen partial pressure, quenched, and analyzed for total zinc, iron, and Fe^{2+}. The typical results are shown in Figure 14, where a kink on an isobar corresponds to the spinel phase boundary. As can be seen from the figure, the calculated isobars are in good agreement with the experimental data, although these data were not used in the optimization. Similarly good agreement was obtained for all other temperatures and oxygen potentials studied in References 15 and 16. It should be noted that the solubility of Fe_2O_3 in the $ZnFe_2O_4$-Fe_3O_4 spinel substantially increases with temperature and reaches about 30 mol pct at 1400 °C.

To the best of our knowledge, there are no reliable data on the cation distributions between octahedral and tetrahedral sites in the Fe_3O_4-$ZnFe_2O_4$ spinel. The predicted site occupancies are given in Figure 15.

Since the liquid-oxide phase appears in the Fe-Zn-O system at rather high temperatures, no data on phase equilibria involving an oxide melt are available, except for those in the Fe-O binary. The parameters of the quasi-chemical model for the liquid oxide were obtained by taking into account

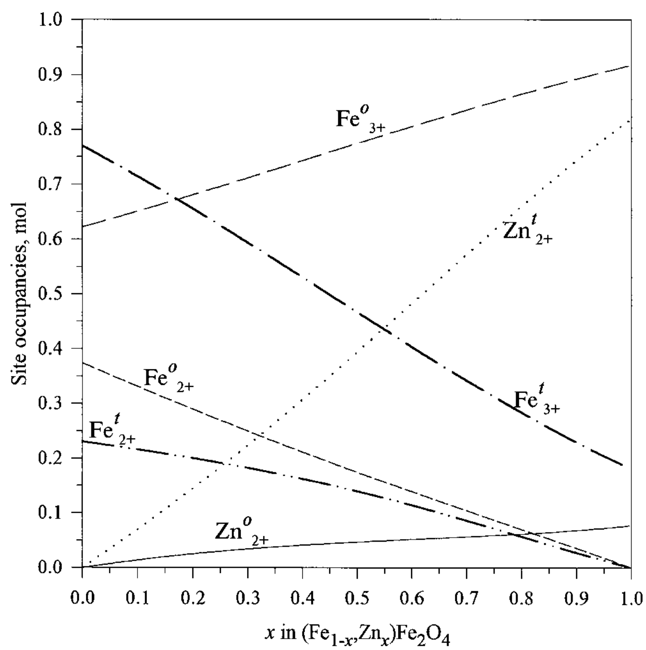


Fig. 15—Predicted octahedral/tetrahedral distributions of cations in $(Fe_{1-x},Zn_x)Fe_2O_4$ spinel at 900 °C.

the phase equilibria in the Fe-Zn-Si-O system. Critical evaluation and optimization of this system are reported elsewhere.[54] The predicted solidus and liquidus in the Fe-Zn-O system in air and at iron saturation are shown in Figures 2 and 10, respectively.

V. CONCLUSIONS

Experimental studies of thermodynamic properties and phase equilibria in the Fe-Zn-O system are subject to errors caused by changes in sample compositions, which result from evaporation of zinc at high temperatures. To resolve some inconsistencies in the literature data, experimental measurements of phase equilibria in the Fe-Zn-O system in air were performed. The phase boundaries were measured by EPMA after equilibration and quenching of the samples. Therefore, the accuracy of these results is not affected by the change of the overall composition due to zinc losses.

In order to apply the CEF to the description of the spinel solid solution $(A,B,E)[A,B,E,V]_2O_4$, the set of model parameters and the sequence of their optimization was proposed. If only a few parameters are needed to reproduce the experimental data, the rest of them are kept equal to zero. In the limiting case, this model reduces to the model for a solution between two simple spinels developed by O'Neill and Navrotsky.[21,22] When the CEF is used to describe the composition dependencies of the critical temperature, magnetic moment, volume, compressibilities, expansivities, *etc.*, it is necessary to introduce many excess terms to obtain the simple linear or quadratic dependencies of these functions as a first approximation. It has been demonstrated how best to assign values to these excess terms and to pseudocomponents based on the experimentally accessible information.

The phase equilibria among the three solid solutions (spinel, wustite, and zincite) and liquid slag dominate in the oxide part of the Fe-Zn-O system. The parameters of the models developed for these solutions were optimized to give self-consistent calculated thermodynamic properties of these phases in the Fe-O and Fe-Zn-O systems. A very large body of experimental information was described with the small number of parameters given in Table II. It should be stressed that the properties of the Fe_3O_4-$ZnFe_2O_4$ solid solution were predicted from those of pure magnetite and zinc ferrite without any additional adjustable parameters. This result lends credence to the model developed for the spinel solution. Although there is no direct experimental evidence of the solubility of Fe_2O_3 in zincite, it was essential to include this compound in the zincite phase in order to reproduce the experimental phase boundaries.

This study is a part of a wider research program, which combines experimental investigation and thermodynamic computer modeling to characterize zinc/lead industrial slags and sinters. The present results have been used at further stages of the program to develop a self-consistent database for the prediction of phase relations and thermodynamic properties of the six-component system PbO-ZnO-CaO-FeO-Fe_2O_3-SiO_2.

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